

Product Requirements Document

Educational Productivity Network (EPN)

Localized Autonomous Agents for Adaptive Curriculum and Study Material Generation

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Table of Contents

Table of Contents.....	2
1. Overview.....	3
1.1 Problem Statement.....	3
1.2 Product Vision	3
1.3 Goals.....	3
1.4 Non-Goals	3
2. Target Users and Personas	3
3. Functional Requirements	4
3.1 Monitor Agent.....	4
3.2 Diagnostic Agent	4
3.3 Curriculum Planner Agent.....	4
3.4 Content Generation Agent.....	4
3.5 Orchestrator Agent	4
3.6 Blockchain / Ledger Layer	4
3.7 Teacher Dashboard.....	5
4. Non-Functional Requirements.....	5
5. Governance and Safety Controls	5
6. Success Metrics.....	6
7. Risks and Mitigations	6
8. Release Phases.....	6
Phase 1 — Pilot (single classroom)	7
Phase 2 — School rollout	7
Phase 3 — District / multi-school.....	7
9. Open Questions	7

1. Overview

1.1 Problem Statement

Teachers cannot continuously monitor every student's individual mastery of every concept, and one-size-fits-all curricula leave both struggling and advanced students underserved. Existing adaptive learning tools typically adjust content difficulty but do not autonomously diagnose root causes, restructure curriculum sequencing, generate new materials, or provide a verifiable, tamper-proof record of student progress and agent decisions.

1.2 Product Vision

EPN is a localized, multi-agent autonomous system that continuously monitors each student's learning progress, diagnoses conceptual gaps, dynamically adapts curriculum sequencing, and generates customized study materials with minimal human intervention — operating under explicit governance controls and producing a blockchain-anchored audit and credentialing trail.

1.3 Goals

- Reduce time-to-intervention for students falling behind on specific concepts.
- Automatically generate targeted, curriculum-aligned study materials at scale.
- Maintain teacher control over high-impact curriculum decisions via HITL approval.
- Provide tamper-evident audit trails for agent actions and student credentials.
- Operate within school data-privacy constraints via localized/edge deployment.

1.4 Non-Goals

- EPN does not replace teachers or remove them from instructional decision-making.
- EPN does not autonomously assign final grades without teacher review.
- EPN is not a public, permissionless blockchain application; the ledger is permissioned and institution-controlled.
- Initial release does not include physical robotics or hardware integration.

2. Target Users and Personas

Persona	Needs	Key interactions
Student	Receive study materials matched to current level; clear progress feedback.	Completes assessments and assigned materials; views progress dashboard.
Teacher	Visibility into class-wide and individual progress; control over curriculum changes; minimal extra workload.	Approves/rejects HITL escalations; reviews summary reports; can override agent decisions.
School administrator / IT	Compliance with data-privacy regulations; manageable infrastructure; auditability.	Configures budgeted-autonomy limits, manages ledger nodes, runs audits.

Parent / guardian	Transparency into child's progress and credentials.	Views progress summaries; optionally verifies credentials on ledger.
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3. Functional Requirements

3.1 Monitor Agent

- FR-1: The system shall ingest quiz/assessment scores, time-on-task, and interaction telemetry (clicks, pauses, retries) from the LMS in near real time.
- FR-2: The system shall detect abnormal error-rate spikes and flag them for circuit-breaker evaluation within one session cycle.

3.2 Diagnostic Agent

- FR-3: The system shall maintain a per-student knowledge graph reflecting mastered concepts, gaps, and misconceptions.
- FR-4: The system shall classify the root cause of repeated errors (e.g., conceptual gap vs. reading-comprehension issue vs. disengagement).

3.3 Curriculum Planner Agent

- FR-5: The system shall generate and update an adaptive task graph per student, sequencing remedial, on-track, and enrichment modules.
- FR-6: Curriculum changes exceeding a configurable impact threshold (e.g., skipping a grade-level unit) shall require teacher approval before activation (HITL).

3.4 Content Generation Agent

- FR-7: The system shall generate study materials (worksheets, quizzes, flashcards, explanations) aligned to the active curriculum plan and student modality preference.
- FR-8: All generated content shall be validated against a curriculum-standards knowledge base prior to delivery.
- FR-9: Each generated content item shall be hashed and the hash recorded on the audit ledger.

3.5 Orchestrator Agent

- FR-10: The system shall enforce budgeted-autonomy limits (daily generation caps, token budgets, re-sequencing frequency) configurable per school/class.
- FR-11: The system shall route any action exceeding policy thresholds to a teacher approval queue and pause related automation until resolved.
- FR-12: The system shall log every autonomous decision (agent, action, timestamp, policy version) to the audit ledger.

3.6 Blockchain / Ledger Layer

- FR-13: The system shall write hashed achievement/credential records, agent action audit entries, and content provenance hashes to a permissioned distributed ledger.

- FR-14: The system shall provide a verification interface allowing teachers, parents, and accrediting institutions to confirm the integrity of a given record against the ledger.
- FR-15: Full student behavioral data and content bodies shall remain off-chain in access-controlled, localized storage; only hashes and minimal metadata are committed on-chain.

3.7 Teacher Dashboard

- FR-16: The system shall provide a dashboard showing class-wide and individual student mastery, pending HITL approvals, and circuit-breaker alerts.
- FR-17: Teachers shall be able to approve, modify, or reject any pending curriculum adjustment or generated content batch.

4. Non-Functional Requirements

Category	Requirement
Privacy	All raw behavioral and assessment data for minors is stored on-premises/edge within the school network; only cryptographic hashes leave the local environment. Retention and purge policies align with FERPA or equivalent regional regulation.
Reliability	Content generation outputs are validated against a curriculum knowledge base; hallucinated or unverifiable content is rejected before delivery (target: 0 unvalidated items reach students).
Latency	Adaptive control loop (Perceive-Reason-Plan-Act-Observe-Adapt) completes within one session cycle (target: under 5 minutes per student per cycle).
Auditability	100% of autonomous agent actions are recorded to the audit ledger with agent ID, action type, timestamp, and policy version.
Scalability	Each localized deployment supports at least one school (up to ~2,000 students) without performance degradation; ledger writes batched to manage throughput.
Accessibility	Generated materials and dashboards meet WCAG 2.1 AA; content modality (text/visual/audio) is configurable per student need.

5. Governance and Safety Controls

These controls map directly to the governance framework requirements (HITL, budgeted autonomy, circuit breakers, policy-as-code) and the blockchain audit layer.

- Human-in-the-Loop: Major curriculum deviations require explicit teacher approval before activation.
- Budgeted autonomy: Per-student/day generation caps and per-agent token/time budgets, configurable by administrators.
- Circuit breakers: Abnormal error-rate spikes pause auto-adaptation and alert a teacher; novel/atypical student profiles trigger HITL review by default.

- Policy-as-code: A centralized gateway validates every generated item against curriculum standards and age-appropriateness filters before delivery.
- Blockchain audit and credentialing: All agent actions, content hashes, and achievement records are committed to a permissioned ledger for tamper-evident verification.

6. Success Metrics

- Reduction in average time between a student developing a concept gap and receiving targeted remediation.
- Percentage improvement in mastery scores on remediated concepts within two weeks of intervention.
- Teacher approval rate and turnaround time for HITL escalations.
- Percentage of generated content passing validation on first pass (target: greater than 98%).
- Number of circuit-breaker events correctly distinguishing disengagement/distress from pure knowledge gaps (validated via teacher feedback).
- Audit completeness: percentage of autonomous actions with a corresponding ledger entry (target: 100%).

7. Risks and Mitigations

Risk	Impact	Mitigation
Hallucinated/incorrect content reaches a student	Misinformation, eroded trust	Curriculum-standards validation gate before delivery; content hashing for traceability
Agent over-adapts curriculum based on noisy data	Student falls further behind or is misclassified	HITL approval for high-impact changes; budgeted-autonomy caps on re-sequencing frequency
Behavioral data exposure	Privacy violation, regulatory non-compliance	Localized/edge storage; only hashes on-chain; access controls and retention limits
Error-rate spike misread as knowledge gap rather than distress	Student wellbeing overlooked	Circuit breaker pauses automation and alerts teacher rather than auto-adjusting
Ledger infrastructure complexity / cost	Delayed rollout, maintenance burden	Use permissioned ledger with managed nodes; batch writes; phase rollout (see Section 8)

8. Release Phases

Phase 1 — Pilot (single classroom)

- Monitor and Diagnostic agents live; manual teacher review of all curriculum suggestions (full HITL).
- Content Generation Agent produces materials in draft form for teacher approval before delivery.
- No ledger; audit log stored in standard database for validation of logging completeness.

Phase 2 — School rollout

- Curriculum Planner Agent operates with HITL thresholds (low-impact changes auto-applied, high-impact require approval).
- Budgeted-autonomy limits and circuit breakers enabled.
- Permissioned ledger introduced for audit log and content-hash recording (read access for administrators).

Phase 3 — District / multi-school

- Achievement/credential records committed to ledger; parent and accrediting-institution verification interface launched.
- Cross-school episodic memory portability for transferring students (privacy-preserving).
- Smart-contract policy gates for credential issuance and cross-institution data sharing.

9. Open Questions

1. Which permissioned ledger platform best fits existing school IT infrastructure and compliance requirements?
2. What is the appropriate default HITL impact threshold, and how should it vary by grade level or subject?
3. How should episodic memory be migrated when a student transfers between EPN-enabled institutions?
4. What is the process for parents/guardians to request review or deletion of their child's behavioral data, and how does this interact with on-chain hashes?